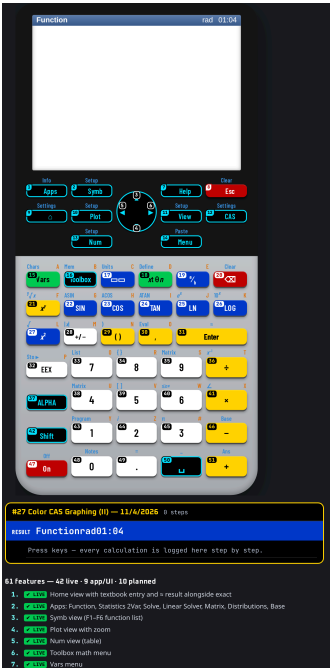
U-45 CALCULATORS

User Manual

Calculator #27 Color CAS Graphing (II)

Graphing - colour



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#27 — Color CAS Graphing (II)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 Apps]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

algebraic with history (type the expression, then EXE / ENTER).

Chapters

1. Getting started
2. Features
3. Numbered key map
4. Key-by-key reference
5. Worked examples
6. Tricks & hacks
7. Keyboard shortcuts
8. Common mistakes
9. Practice drills
10. Appendix A — Drill answers
11. Appendix B — Function finder (A–Z)

12. Appendix C — Glossary

13. Appendix D — Model notes & limits

14. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#27** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#28, #32, #33, #34, #35, #38, #39, #40, #43, #44, #45, #48, #49
op	Operators & execute	#21, #29, #30, #31, #36, #41, #46, #51
fn	Functions	#17, #19, #22, #23, #24, #25, #26, #27
mem	Memory, variables & registers	#15, #18
clr	Clear, delete & power	#8, #20, #47

mod	**Shift, modes, menus & navigation**	**#1, #2, #3, #4, #5, #6, #7, #9, #10, #11, #12, #13, #14, #16, #37, #42, #50**

2 · Features

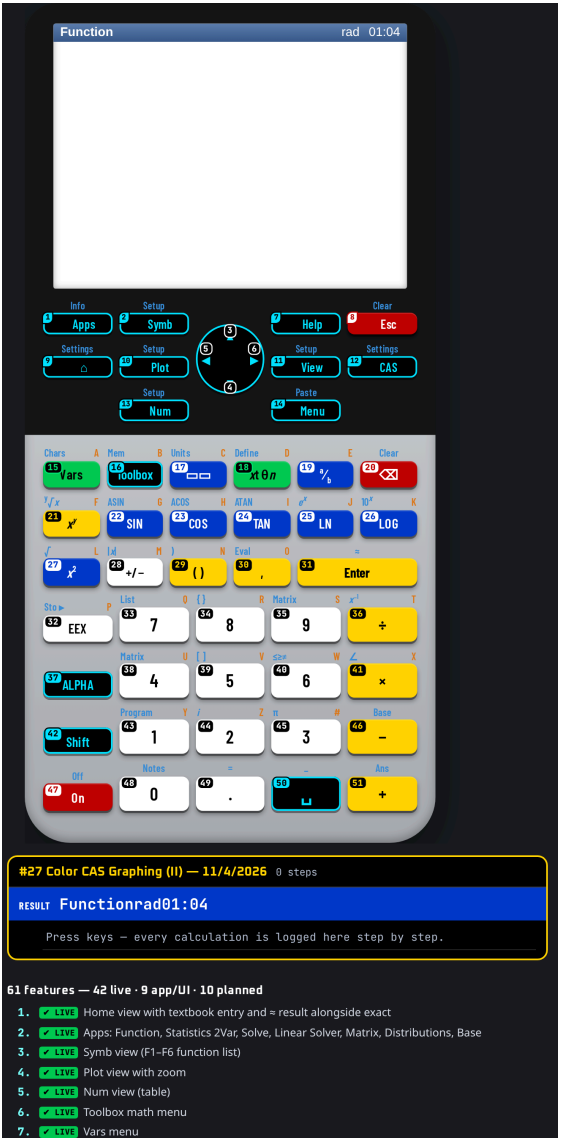
61 features: 42 live on the keys, 9 app/UI features, 10 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Home view with textbook entry and ≈ result alongside exact**
2. LIVE **Apps: Function, Statistics 2Var, Solve, Linear Solver, Matrix, Distributions, Base**
3. LIVE **Symb view (F1–F6 function list)**
4. LIVE **Plot view with zoom**
5. LIVE **Num view (table)**
6. LIVE **Toolbox math menu**
7. LIVE **Vars menu**
8. LIVE **Variables A–Z**
9. LIVE **Fraction template ▭/▭ with exact fraction results**
10. LIVE **S⇔D / F⇔D toggle between exact (fraction, √, π) and decimal answers**
11. LIVE **Square root √**
12. LIVE **Any-index root $\sqrt[x]{y}$**
13. LIVE **Square x^2**
14. LIVE **Reciprocal x^{-1} (1/x)**
15. LIVE **Power x^y / ^ with right-to-left chaining (2^3^2 = 512)**
16. LIVE **Common logarithm $\log_{10}$**
17. LIVE **Natural logarithm ln**
18. LIVE **Antilog 10^x**
19. LIVE **Exponential e^x**

20. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
21. LIVE **Inverse trigonometry $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$ with domain checking (Math ERROR outside −1…1)**
22. LIVE **Absolute value |x|**
23. LIVE **π constant to full internal precision**
24. LIVE **Scientific-notation entry ×10^x / EXP / EEX**
25. LIVE **Sign change (−) / +/− / CHS**
26. LIVE **Ans recall of the last result; operators auto-insert Ans**
27. LIVE **Store a value into memory / a variable (STO, sto▸, →, Min)**
28. LIVE **Named variables (A–F, X, Y, M … or A–Z) usable inside expressions**
29. LIVE **SOLVE — numeric equation solver (Newton / bisection)**
30. LIVE **TABLE — tabulate f(x) over start / end / step**
31. LIVE **Statistics: data entry, n, $\bar{x}$, σ, s, Σx, Σx^2**
32. LIVE **Two-variable statistics and linear regression (a, b, r)**
33. LIVE **Matrix mode: define, add, multiply, det, transpose, identity**
34. LIVE **Complex numbers: i, arg, conjugate, Re / Im, polar ↔ rectangular**
35. LIVE **BASE-N: DEC / HEX / BIN / OCT conversions**
36. LIVE **Bitwise logic AND / OR / XOR / XNOR / NOT**
37. LIVE **Unit conversions (length, mass, volume, temperature)**
38. LIVE **Backspace / DEL editing of the current entry**
39. LIVE **Replay ▲ ▼ — scroll back to previous calculations and re-edit them**
40. LIVE **CATALOG — alphabetical list of every function**
41. LIVE **MATH / Toolbox menu of templates and functions**
42. LIVE **Probability distributions (normal, binomial, Poisson) and PRB menu**

43. APP/UI **Every key numbered 1…N in reading order — matches the key map in the user manual**
44. APP/UI **High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle**
45. APP/UI **Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)**
46. APP/UI **Show / hide key numbers with one button (remembered between sessions)**
47. APP/UI **Copy the current display text to the clipboard**
48. APP/UI **Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i**
49. APP/UI **Fit / 1:1 zoom that never pushes keys off a phone screen**
50. APP/UI **Reference photo viewer and tool-tip name on every key**
51. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
52. PLANNED **CAS view**
53. PLANNED **Help system**
54. PLANNED **Notes**
55. PLANNED **Programming**
56. PLANNED **Geometry app**
57. PLANNED **Graph image export (PNG)**
58. PLANNED **Shareable function list link**
59. PLANNED **Dark screen theme**
60. PLANNED **Speech read-out of values**
61. PLANNED **Exam mode**

3 · Numbered key map



Calculator #27 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	Apps	Info	·	·	mod
2	Symb	Setup	·	·	mod
3	▲	·	·	·	mod
4	▼	·	·	·	mod
5	◀	·	·	·	mod
6	▶	·	·	·	mod
7	Help	·	·	·	mod
8	Esc	Clear	·	·	clr
9	⌂	Settings	·	·	mod
10	Plot	Setup	·	·	mod
11	View	Setup	·	·	mod
12	CAS	Settings	·	·	mod
13	Num	Setup	·	·	mod
14	Menu	Paste	·	·	mod
15	Vars	Chars	A	·	mem
16	Toolbox	Mem	B	·	mod
17	▭▭	Units	C	·	fn
18	xt θn	Define	D	·	mem
19	a⁄b	·	E	·	fn
20	⌫	Clear	·	·	clr
21	xy	y√x	F	·	op
22	SIN	ASIN	G	·	fn
23	COS	ACOS	H	·	fn

#	Face	Shift	Alpha	3rd	Group
24	TAN	ATAN	I	·	fn
25	LN	ex	J	·	fn
26	LOG	10x	K	·	fn
27	x^2	√	L	·	fn
28	+/−	|x|	M	·	num
29	( )	)	N	·	op
30	,	Eval	O	·	op
31	Enter	≈	·	·	op
32	EEX	Sto ▶	P	·	num
33	7	List	Q	·	num
34	8	{ }	R	·	num
35	9	Matrix	S	·	num
36	÷	x^{-1}	T	·	op
37	ALPHA	·	·	·	mod
38	4	Matrix	U	·	num
39	5	[ ]	V	·	num
40	6	≤≥≠	W	·	num
41	×	∠	X	·	op
42	Shift	·	·	·	mod
43	1	Program	Y	·	num
44	2	i	Z	·	num
45	3	π	#	·	num
46	−	Base	·	·	op
47	On	Off	·	·	clr
48	0	Notes	·	·	num
49	.	=	·	·	num
50	␣	_	·	·	mod
51	+	Ans	·	·	op

4 · Key-by-key reference

Every key on calculator #27, in number order, with every function it carries in this app.

#1 Apps `Shift, modes, menus & navigation`

Primary: APPS — applications

shifted (2nd / f / SHIFT layer) (press #42 first): Info — Shows the message "Info" (feature not provided in this app)

#2 Symb `Shift, modes, menus & navigation`

Primary: Symbolic view

shifted (2nd / f / SHIFT layer) (press #42 first): Setup — TBLSET — table start and step

#3 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#4 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#5 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#6 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#7 Help `Shift, modes, menus & navigation`

Primary: Shows the message “Help is not available” (feature not provided in this app)

#8 Esc `Clear, delete & power`

Primary: QUIT — leave the current screen / app

shifted (2nd / f / SHIFT layer) (press #42 first): Clear — CLEAR — clear the entry line / screen

#9 ⌂ `Shift, modes, menus & navigation`

Primary: HOME — return to the main calculation screen

shifted (2nd / f / SHIFT layer) (press #42 first): Settings — Graph / app mode

#10 Plot `Shift, modes, menus & navigation`

Primary: Plot view

shifted (2nd / f / SHIFT layer) (press #42 first): Setup — WINDOW — set the graph view

#11 View `Shift, modes, menus & navigation`

Primary: ZOOM — zoom presets

shifted (2nd / f / SHIFT layer) (press #42 first): Setup — WINDOW — set the graph view

#12 CAS `Shift, modes, menus & navigation`

Primary: Shows the message “CAS not available: numeric Home only” (feature not provided in this app)

shifted (2nd / f / SHIFT layer) (press #42 first): Settings — Graph / app mode

#13 Num `Shift, modes, menus & navigation`

Primary: Numeric view

shifted (2nd / f / SHIFT layer) (press #42 first): Setup — TBLSET — table start and step

#14 Menu `Shift, modes, menus & navigation`

Primary: MATH menu — extra math functions

shifted (2nd / f / SHIFT layer) (press #42 first): Paste — ENTRY — recall a previous entry

#15 Vars `Memory, variables & registers`

Primary: VARS — variables menu

shifted (2nd / f / SHIFT layer) (press #42 first): Chars — CATALOG — list of all commands

alpha layer (press #37 first): A — Variable A (store with STO/ →, recall by name)

#16 Toolbox `Shift, modes, menus & navigation`

Primary: MATH menu — extra math functions

shifted (2nd / f / SHIFT layer) (press #42 first): Mem — RESET — return to factory defaults

alpha layer (press #37 first): B — Variable B (store with STO/ →, recall by name)

#17 ▭▭ `Functions`

Primary: Fraction template ▭/▭ (numerator over denominator)

shifted (2nd / f / SHIFT layer) (press #42 first): Units — Conversion menu

alpha layer (press #37 first): C — Variable C (store with STO/ →, recall by name)

#18 xt θn `Memory, variables & registers`

Primary: The graphing variable (X / x,θ,t,n) for Y= functions, tables and solving

shifted (2nd / f / SHIFT layer) (press #42 first): Define — Y= — enter functions to graph

alpha layer (press #37 first): D — Variable D (store with STO/ →, recall by name)

#19 a/b Functions

Primary: S⇔D — toggle the result between exact (fraction, √, π) and decimal

alpha layer (press #37 first): E — Variable E (store with STO/ →, recall by name)

#20 ⌫ Clear, delete & power

Primary: DEL — delete the character at the cursor

shifted (2nd / f / SHIFT layer) (press #42 first): Clear — CLEAR — clear the entry line / screen

Keyboard: Backspace Delete

#21 xy Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #42 first): $y\sqrt{}x$ — n-th root $\sqrt[x]{y}$

alpha layer (press #37 first): F — Variable F (store with STO/ →, recall by name)

Keyboard: ^

#22 SIN Functions

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #42 first): ASIN — Inverse sine $\sin^{-1}$

alpha layer (press #37 first): G — Variable G (store with STO/ →, recall by name)

Keyboard: s

#23 COS Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #42 first): ACOS — Inverse cosine $\cos^{-1}$

alpha layer (press #37 first): H — Variable H (store with STO/ →, recall by name)

Keyboard: c

#24 TAN Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #42 first): ATAN — Inverse tangent $\tan^{-1}$

alpha layer (press #37 first): I — Variable I (store with STO/ →, recall by name)

Keyboard: t

#25 LN Functions

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #42 first): ex — e^x — exponential

alpha layer (press #37 first): J — Variable J (store with STO/ →, recall by name)

Keyboard: n

#26 LOG Functions

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #42 first): 10x — 10^x — antilog

alpha layer (press #37 first): K — Variable K (store with STO/ →, recall by name)

Keyboard: l

#27 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #42 first): $\sqrt{\ }$ — Square root $\sqrt{\ }$

alpha layer (press #37 first): L — Variable L (store with STO/ →, recall by name)

Keyboard: q

#28 +/− Digits & number entry

Primary: Negative sign (−) — makes the next number negative. Different from the subtraction key

shifted (2nd / f / SHIFT layer) (press #42 first): |x| — Absolute value |x|

alpha layer (press #37 first): M — Variable M (store with STO/ →, recall by name)

#29 ( ) Operators & execute

Primary: Opening bracket (

shifted (2nd / f / SHIFT layer) (press #42 first):) — Closing bracket). Closing brackets at the end of a line may be omitted

alpha layer (press #37 first): N — Variable N (store with STO/ →, recall by name)

Keyboard: (

#30 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

shifted (2nd / f / SHIFT layer) (press #42 first): Eval — Execute / = — evaluate the line and show the result (EXE / ENTER / =)

alpha layer (press #37 first): 0 — Variable 0 (store with STO/ →, recall by name)

Keyboard: [,]

#31 Enter Operators & execute

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

shifted (2nd / f / SHIFT layer) (press #42 first): ≈ — ▶Decimal — show the result as a decimal

Keyboard: [Enter] [=]

#32 EEX Digits & number entry

Primary: ×10^{x} / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^{x} 23

shifted (2nd / f / SHIFT layer) (press #42 first): Sto ▸ — STO▶ (→) — store into a variable

alpha layer (press #37 first): P — Variable P (store with STO/ →, recall by name)

Keyboard: [e]

#33 7 Digits & number entry

Primary: Digit 7

shifted (2nd / f / SHIFT layer) (press #42 first): List — LIST menu

alpha layer (press #37 first): Q — Variable Q (store with STO/ →, recall by name)

Keyboard: 7

#34 8 Digits & number entry

Primary: Digit 8

shifted (2nd / f / SHIFT layer) (press #42 first): { } — Types "{"

alpha layer (press #37 first): R — Variable R (store with STO/ →, recall by name)

Keyboard: 8

#35 9 Digits & number entry

Primary: Digit 9

shifted (2nd / f / SHIFT layer) (press #42 first): Matrix — MATRIX menu

alpha layer (press #37 first): S — Variable S (store with STO/ →, recall by name)

Keyboard: 9

#36 ÷ Operators & execute

Primary: Division ÷

shifted (2nd / f / SHIFT layer) (press #42 first): x^{-1} — Reciprocal x^{-1} (1/x)

alpha layer (press #37 first): T — Variable T (store with STO/ →, recall by name)

Keyboard: /

#37 ALPHA `Shift, modes, menus & navigation`

Primary: ALPHA — the next key types its letter/variable (red/green label)

#38 4 `Digits & number entry`

Primary: Digit 4

shifted (2nd / f / SHIFT layer) (press #42 first): Matrix — MATRIX menu

alpha layer (press #37 first): U — Variable U (store with STO/ →, recall by name)

Keyboard: 4

#39 5 `Digits & number entry`

Primary: Digit 5

shifted (2nd / f / SHIFT layer) (press #42 first): [ ] — Types "["

alpha layer (press #37 first): V — Variable V (store with STO/ →, recall by name)

Keyboard: 5

#40 6 `Digits & number entry`

Primary: Digit 6

shifted (2nd / f / SHIFT layer) (press #42 first): ≤≥≠ — Types "≠"

alpha layer (press #37 first): W — Variable W (store with STO/ →, recall by name)

Keyboard: 6

#41 × Operators & execute

Primary: Multiplication ×

alpha layer (press #37 first): X — Variable X (store with STO/ →, recall by name)

Keyboard: * x

#42 Shift Shift, modes, menus & navigation

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#43 1 Digits & number entry

Primary: Digit 1

shifted (2nd / f / SHIFT layer) (press #42 first): Program — PRGM — programs

alpha layer (press #37 first): Y — Variable Y (store with STO/ →, recall by name)

Keyboard: 1

#44 2 Digits & number entry

Primary: Digit 2

shifted (2nd / f / SHIFT layer) (press #42 first): i — Imaginary unit i (complex numbers)

alpha layer (press #37 first): Z — Variable Z (store with STO/ →, recall by name)

Keyboard: 2

#45 3 `Digits & number entry`

Primary: Digit 3

shifted (2nd / f / SHIFT layer) (press #42 first): π — π (3.14159265…) as an exact constant

Keyboard: `3`

#46 – `Operators & execute`

Primary: Subtraction –

shifted (2nd / f / SHIFT layer) (press #42 first): Base — BASE-N — number-base menu (DEC/HEX/BIN/OCT)

Keyboard: `-`

#47 On `Clear, delete & power`

Primary: ON / AC — power on, or clear all

shifted (2nd / f / SHIFT layer) (press #42 first): Off — OFF — power off (state is lost; press ON to restart)

#48 0 `Digits & number entry`

Primary: Digit 0

shifted (2nd / f / SHIFT layer) (press #42 first): Notes — Shows the message "Notes not available" (feature not provided in this app)

Keyboard: `0`

#49 . Digits & number entry

Primary: Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

shifted (2nd / f / SHIFT layer) (press #42 first): = — Equals — complete the calculation

Keyboard: .

#50 ␣ Shift, modes, menus & navigation

Primary: Printed label only (no action in this app)

#51 + Operators & execute

Primary: Addition +

shifted (2nd / f / SHIFT layer) (press #42 first): Ans — Ans — recall the last answer into the current expression

Keyboard: +

5 · Worked examples

18 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

#43 1 → #44 2 → #51 + → #45 3 → #41 × → #38 4 → #42 Shift → #30 Eval

Display shows 24 **— expected 24. ✔ verified on the app.**

Multiplication is done before addition, so the calculator holds the + until the × part is finished.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

#29 ( ) → #43 1 → #44 2 → #51 + → #34 8 → #42 Shift → #29) → #36 ÷ → #38 4 → #42 Shift → #30 Eval

Display shows 5 **— expected 5. ✔ verified on the app.**

Brackets force the sum to be worked out before the division.

Example 5.3 — Two brackets multiplied

Calculate **(3 + 4) × (9 − 2)**. Press **Reset** first.

#29 ( ) → #45 3 → #51 + → #38 4 → #42 Shift → #29) → #41 × → #29 ( ) → #35 9 → #46 − → #44 2 → #42 Shift → #29) → #42 Shift → #30 Eval

Display shows 49 — expected 49. ✔ verified on the app.

Both brackets are evaluated first, then multiplied.

Example 5.4 — Squares

Calculate 12^2. Press **Reset** first.

#43 1 → #44 2 → #27 x^2 → #42 Shift → #30 Eval

Display shows 144 — expected 144. ✔ verified on the app.

x^2 squares the value — quicker and safer than typing the number twice.

Example 5.5 — Square roots

Calculate $\sqrt{144}$. Press **Reset** first.

#42 Shift → #27 √ → #43 1 → #38 4 → #38 4 → #42 Shift → #30 Eval

Display shows 12 — expected 12. ✔ verified on the app.

The square-root key returns the principal (positive) root.

Example 5.6 — Powers

Calculate 2^10. Press Reset first.

#44 2 → #21 x^y → #43 1 → #48 0 → #42 Shift → #30 Eval

Display shows 1024 **— expected 1024. ✔ verified on the app.**

The power key raises the first number to the second.

Example 5.7 — Reciprocals

Calculate 1 ÷ 0.25 (x^{-1}). Press Reset first.

#48 0 → #49 . → #44 2 → #39 5 → #42 Shift → #36 x^{-1} → #42 Shift → #30 Eval

Display shows 4 **— expected 4. ✔ verified on the app.**

x^{-1} (1/x) gives the reciprocal — handy for resistors in parallel or rates.

Example 5.8 — Trigonometry (degrees)

Calculate cos 0° (Deg mode). Press Reset first.

#23 COS → #48 0 → #42 Shift → #29) → #42 Shift → #30 Eval

Display shows 1 **— expected 1. ✔ verified on the app.**

Trig functions use the current angle unit; these answers assume degrees (the start-up setting).

Example 5.9 — Common logarithm

Calculate **log 1000**. Press **Reset** first.

#26 LOG → #43 1 → #48 0 → #48 0 → #48 0 → #42 Shift → #29) → #42 Shift → #30 Eval

Display shows 3 — expected 3. ✔ verified on the app.

log is base-10: it counts the zeros of a power of ten.

Example 5.10 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

#25 LN → #43 1 → #42 Shift → #29) → #42 Shift → #30 Eval

Display shows 0 — expected 0. ✔ verified on the app.

ln is the natural (base-e) logarithm.

Example 5.11 — Negative numbers

Calculate **−5 + 12**. Press **Reset** first.

#28 +/− → #39 5 → #51 + → #43 1 → #44 2 → #42 Shift → #30 Eval

Display shows 7 — expected 7. ✔ verified on the app.

Use the sign key for a negative number — not the subtraction key.

Example 5.12 — Scientific notation

Calculate 6×10^2 × 3. Press Reset first.

#40 6 → #32 EEX → #44 2 → #41 × → #45 3 → #42 Shift → #30 Eval

Display shows 1800 **— expected 1800. ✔ verified on the app.**

The exponent key enters ×10ⁿ without typing × 10 ^.

Example 5.13 — Area of a circle πr²

Calculate π × 2.5². Press Reset first.

#42 Shift → #45 π → #41 × → #44 2 → #49 . → #39 5 → #27 x^2 → #42 Shift → #30 Eval

Display shows 19.63495408 **— expected 19.63495408. ✔ verified on the app.**

π is stored to full precision — never type 3.14.

Example 5.14 — Continue from the answer

Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.

#43 1 → #48 0 → #48 0 → #41 × → #38 4 → #42 Shift → #30 Eval → #36 ÷ → #44 2 → #42 Shift → #30 Eval

Display shows 200 **— expected 200. ✔ verified on the app.**

Pressing an operator straight after a result continues from that answer (Ans).

Example 5.15 — Average of three numbers

Calculate (4 + 8 + 15) ÷ 3. Press Reset first.

#29 ( ) → #38 4 → #51 + → #34 8 → #51 + → #43 1 → #39 5 → #42 Shift → #29) → #36 ÷ → #45 3 → #42 Shift → #30 Eval

Display shows `9` **— expected 9. ✔ verified on the app.**

Add up the values, then divide by how many there are.

Example 5.16 — Mixed operations

Calculate 50 − 4 × 6 + 7. Press Reset first.

#39 5 → #48 0 → #46 − → #38 4 → #41 × → #40 6 → #51 + → #33 7 → #42 Shift → #30 Eval

Display shows `33` **— expected 33. ✔ verified on the app.**

Only the × part is done first; the − and + are then worked left to right.

Example 5.17 — Dividing twice

Calculate 360 ÷ 4 ÷ 9. Press Reset first.

#45 3 → #40 6 → #48 0 → #36 ÷ → #38 4 → #36 ÷ → #35 9 → #42 Shift → #30 Eval

Display shows `10` **— expected 10. ✔ verified on the app.**

Division is worked left to right: the first quotient is divided again.

Example 5.18 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.

#29 ( ) → #29 ( ) → #45 3 → #51 + → #39 5 → #42 Shift →
#29) → #41 × → #38 4 → #46 − → #44 2 → #42 Shift →
#29) → #36 ÷ → #43 1 → #48 0 → #42 Shift → #30 Eval

Display shows 3 — expected 3. ✔ verified on the app.

The innermost brackets are worked out first, then the outer brackets, then the division.

6 · Tricks & hacks

Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 16: Hidden functions on this model

67 extra functions sit on shifted layers. Press the layer key once, then the key: for example #42 then #1 → Info; #42 then #2 → Setup; #42 then #8 → Clear; #42 then #9 → Settings.

Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #51.

Hack 18: Type faster on a computer

27 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
Backspace Delete	#20	⌫	DEL — delete the character at the cursor
^	#21	xy	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
s	#22	SIN	Sine (uses the current angle unit)
c	#23	COS	Cosine (current angle unit)
t	#24	TAN	Tangent (current angle unit)
n	#25	LN	Natural logarithm ln
l	#26	LOG	Common logarithm $\log_{10}$
q	#27	x^2	Square x^2
(	#29	( )	Opening bracket (
,	#30	,	Comma / argument separator (between function arguments, list items or coordinates)
Enter =	#31	Enter	Execute / = — evaluate the line and show the result (EXE / ENTER / =)
e	#32	EEX	×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23
7	#33	7	Digit 7
8	#34	8	Digit 8
9	#35	9	Digit 9
/	#36	÷	Division ÷
4	#38	4	Digit 4
5	#39	5	Digit 5
6	#40	6	Digit 6

* X	#41	×	Multiplication ×
1	#43	1	Digit 1
2	#44	2	Digit 2
3	#45	3	Digit 3
-	#46	–	Subtraction –
0	#48	0	Digit 0
.	#49	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
+	#51	+	Addition +

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: (–) vs –

Use the negative key for negative numbers.

Mistake 2: Radian mode

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

Mistake 3: Implicit multiplication

2/3x may be read as 2/(3x) depending on the model; use brackets.

Mistake 4: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 5: Wrong angle unit

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 6: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (92)

Work each one on calculator #27, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (250 + 350) ÷ 12 = _____
14. (5 + 6) × (10 − 3) = _____
15. (12 + 3) × (8 − 5) = _____
16. (1.5 + 2.5) × (7 − 4) = _____

17. (20 + 30) × (15 − 5) = _____

18. (9 + 1) × (100 − 1) = _____

19. (11 + 13) × (17 − 9) = _____

20. 25^2 = _____

21. 7^2 = _____

22. 1.5^2 = _____

23. 19^2 = _____

24. 33^2 = _____

25. 0.5^2 = _____

26. 125^2 = _____

27. 64^2 = _____

28. √625 = _____

29. √1024 = _____

30. √9801 = _____

31. 3^5 = _____

32. 1.05^3 = _____

33. 5^4 = _____

34. 7^3 = _____

35. 1.5^2 = _____

36. 10^6 = _____

37. 2^0.5 = _____

38. 1 ÷ 0.2 (x^{-1}) = _____

39. log 100 = _____

40. log 10000 = _____

41. ln 10 = _____

42. ln 2 = _____

43. −20 + 7 = _____

44. −3 + 3 = _____

45. −100 + 250 = _____

46. −8 + 1 = _____

47. 2×10^3 × 4 = _____

48. 1.5×10^4 × 2 = _____

49. 3×10^5 × 3 = _____

50. 4.5×10^2 × 2 = _____

51. π × 1.2^2 = _____

52. 15 × 6 = , then Ans ÷ 9 = _____

53. 7.5 × 8 = , then Ans ÷ 3 = _____

54. 250 × 12 = , then Ans ÷ 100 = _____

55. (12 + 15 + 18) ÷ 3 = _____

56. (70 + 85 + 91) ÷ 3 = _____

57. (2.5 + 3.5 + 6) ÷ 3 = _____

58. (100 + 200 + 600) ÷ 3 = _____

59. 100 − 3 × 12 + 9 = _____

60. 18 − 2 × 5 + 40 = _____

61. 7.5 − 1.5 × 3 + 2 = _____

62. 64 − 8 × 4 + 10 = _____

63. 300 − 25 × 4 + 55 = _____

64. 9 − 3 × 3 + 1 = _____

65. 45 − 5 × 5 + 15 = _____

66. 1000 ÷ 8 ÷ 5 = _____

67. 96 ÷ 2 ÷ 3 = _____

68. 7200 ÷ 60 ÷ 60 = _____

69. 500 ÷ 4 ÷ 25 = _____

70. 81 ÷ 3 ÷ 3 = _____

71. ((10 + 2) × 3 − 6) ÷ 6 = _____

72. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

73. ((20 + 30) × 2 − 25) ÷ 3 = _____

74. ((7 + 8) × 9 − 15) ÷ 12 = _____

75. 12 + 3 × 4 = _____

76. (12 + 8) ÷ 4 = _____

77. (3 + 4) × (9 − 2) = _____

78. 12^2 = _____

79. √144 = _____

80. 2^10 = _____

81. 1 ÷ 0.25 (x^{-1}) = _____

82. cos 0° (Deg mode) = _____

83. log 1000 = _____

84. ln 1 = _____

85. −5 + 12 = _____

86. 6×10^2 × 3 = _____

87. π × 2.5^2 = _____

88. 100 × 4 = , then Ans ÷ 2 = _____

89. (4 + 8 + 15) ÷ 3 = _____

90. 50 − 4 × 6 + 7 = _____

91. 360 ÷ 4 ÷ 9 = _____

92. ((3 + 5) × 4 − 2) ÷ 10 = _____

Set B — Key-finding drills (107)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “Apps” (APPS)? _____
2. Which key is “Symb” (Symbolic view)? _____

3. **Which key is "▲" (Cursor ▲)? _____**
4. **Which key is "▼" (Cursor ▼)? _____**
5. **Which key is "◀" (Cursor ◀)? _____**
6. **Which key is "▶" (Cursor ▶)? _____**
7. **Which key is "Help" (Shows the message "Help is not available")? _____**
8. **Which key is "Esc" (QUIT)? _____**
9. **Which key is "⌂" (HOME)? _____**
10. **Which key is "Plot" (Plot view)? _____**
11. **Which key is "View" (ZOOM)? _____**
12. **Which key is "CAS" (Shows the message "CAS not available: numeric Home only")? _____**
13. **Which key is "Num" (Numeric view)? _____**
14. **Which key is "Menu" (MATH menu)? _____**
15. **Which key is "Vars" (VARS)? _____**
16. **Which key is "Toolbox" (MATH menu)? _____**
17. **Which key is "▭▭" (Fraction template ▭/▭)? _____**
18. **Which key is "xt θn" (The graphing variable)? _____**
19. **Which key is "a⁄b" (S⇔D)? _____**
20. **Which key is "⌫" (DEL)? _____**
21. **Which key is "xy" (Power x^y / ^)? _____**
22. **Which key is "SIN" (Sine)? _____**

23. Which key is “COS” (Cosine)? _____

24. Which key is “TAN” (Tangent)? _____

25. Which key is “LN” (Natural logarithm ln)? _____

26. Which key is “LOG” (Common logarithm $\log_{10}$)? _____

27. Which key is “x^2” (Square x^2)? _____

28. Which key is “+/–” (Negative sign)? _____

29. Which key is “( )” (Opening bracket)? _____

30. Which key is “,” (Comma / argument separator)? _____

31. Which key is “Enter” (Execute / =)? _____

32. Which key is “EEX” (×10^x / EXP / EE)? _____

33. Which key is “÷” (Division ÷)? _____

34. Which key is “ALPHA” (ALPHA)? _____

35. Which key is “×” (Multiplication ×)? _____

36. Which key is “Shift” (SHIFT / 2nd)? _____

37. Which key is “–” (Subtraction –)? _____

38. Which key is “On” (ON / AC)? _____

39. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

40. Which key is “+” (Addition +)? _____

41. How do you reach “Info”? _____

42. How do you reach “Setup”? _____

43. How do you reach “Clear”? _____

44. How do you reach “Settings”? _____

45. How do you reach “Setup”? _____

46. How do you reach “Setup”? _____

47. How do you reach “Settings”? _____

48. How do you reach “Setup”? _____

49. How do you reach “Paste”? _____

50. How do you reach “Chars”? _____

51. How do you reach “A”? _____

52. How do you reach “Mem”? _____

53. How do you reach “B”? _____

54. How do you reach “Units”? _____

55. How do you reach “C”? _____

56. How do you reach “Define”? _____

57. How do you reach “D”? _____

58. How do you reach “E”? _____

59. How do you reach “Clear”? _____

60. How do you reach “$y\sqrt{x}$”? _____

61. How do you reach “F”? _____

62. How do you reach “ASIN”? _____

63. How do you reach “G”? _____

64. How do you reach “ACOS”? _____

65. How do you reach “H”? _____

66. How do you reach “ATAN”? _____

67. How do you reach “I”? _____

68. How do you reach “ex”? _____

69. How do you reach “J”? _____

70. How do you reach “10x”? _____

71. How do you reach “K”? _____

72. How do you reach “√”? _____

73. How do you reach “L”? _____

74. How do you reach “|x|”? _____

75. How do you reach “M”? _____

76. How do you reach “)”? _____

77. How do you reach “N”? _____

78. How do you reach “Eval”? _____

79. How do you reach “0”? _____

80. How do you reach “≈”? _____

81. How do you reach “Sto ▸”? _____

82. How do you reach “P”? _____

83. How do you reach “List”? _____

84. How do you reach “Q”? _____

85. How do you reach “{ }”? _____

86. How do you reach “R”? _____

87. How do you reach “Matrix”? _____

88. How do you reach “S”? _____

89. How do you reach “x^{-1}”? _____

90. How do you reach “T”? _____

91. How do you reach “Matrix”? _____

92. How do you reach “U”? _____

93. How do you reach “[ ]”? _____

94. How do you reach “V”? _____

95. How do you reach “≤≥≠”? _____

96. How do you reach “W”? _____

97. How do you reach “X”? _____

98. How do you reach “Program”? _____

99. How do you reach “Y”? _____

100. How do you reach “i”? _____

101. How do you reach “Z”? _____

102. How do you reach “π”? _____

103. How do you reach “Base”? _____

104. How do you reach “Off”? _____

105. How do you reach “Notes”? _____

106. How do you reach “=”? _____

107. How do you reach “Ans”? _____

Appendix A — Drill answers

Set A

1. **55** (display `55`)

   `#44` 2 → `#39` 5 → `#51` + → `#39` 5 → `#41` × → `#40` 6 → `#42` Shift → `#30` Eval

2. **79** (display `79`)

   `#33` 7 → `#51` + → `#34` 8 → `#41` × → `#35` 9 → `#42` Shift → `#30` Eval

3. **282** (display `282`)

   `#43` 1 → `#39` 5 → `#48` 0 → `#51` + → `#43` 1 → `#44` 2 → `#41` × → `#43` 1 → `#43` 1 → `#42` Shift → `#30` Eval

4. **213** (display `213`)

   `#45` 3 → `#51` + → `#43` 1 → `#38` 4 → `#41` × → `#43` 1 → `#39` 5 → `#42` Shift → `#30` Eval

5. **90** (display `90`)

   `#38` 4 → `#34` 8 → `#51` + → `#40` 6 → `#41` × → `#33` 7 → `#42` Shift → `#30` Eval

6. **1100** (display `1100`)

   `#43` 1 → `#48` 0 → `#48` 0 → `#48` 0 → `#51` + → `#44` 2 → `#39` 5 → `#41` × → `#38` 4 → `#42` Shift → `#30` Eval

7. 82 (display `82`)

#43 1 → #35 9 → #51 + → #44 2 → #43 1 → #41 × → #45 3 → #42 Shift → #30 Eval

8. 128 (display `128`)

#40 6 → #38 4 → #51 + → #34 8 → #41 × → #34 8 → #42 Shift → #30 Eval

9. 10 (display `10`)

#29 ( ) → #38 4 → #39 5 → #51 + → #43 1 → #39 5 → #42 Shift → #29) → #36 ÷ → #40 6 → #42 Shift → #30 Eval

10. 8 (display `8`)

#29 ( ) → #35 9 → #51 + → #33 7 → #42 Shift → #29) → #36 ÷ → #44 2 → #42 Shift → #30 Eval

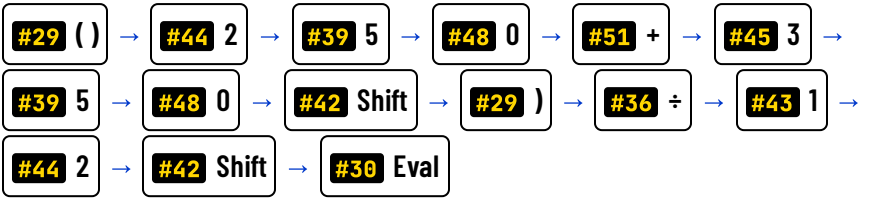
11. 20 (display `20`)

#29 ( ) → #43 1 → #48 0 → #48 0 → #51 + → #40 6 → #48 0 → #42 Shift → #29) → #36 ÷ → #34 8 → #42 Shift → #30 Eval

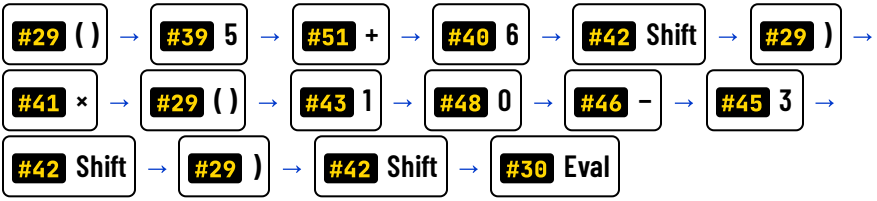
12. 10 (display `10`)

#29 ( ) → #45 3 → #45 3 → #51 + → #43 1 → #33 7 → #42 Shift → #29) → #36 ÷ → #39 5 → #42 Shift → #30 Eval

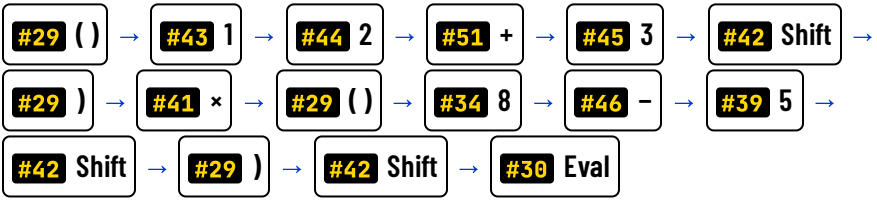
13. 50 (display `50`)



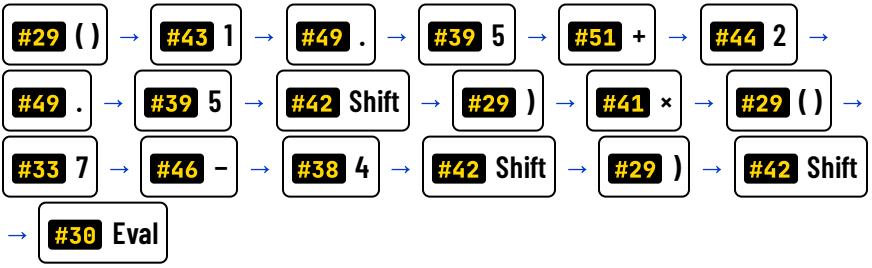
14. 77 (display `77`)



15. 45 (display `45`)



16. 12 (display `12`)



17. 500 (display `500`)

#29 ( ) → #44 2 → #48 0 → #51 + → #45 3 → #48 0 → #42 Shift → #29) → #41 × → #29 ( ) → #43 1 → #39 5 → #46 − → #39 5 → #42 Shift → #29) → #42 Shift → #30 Eval

18. 990 (display `990`)

#29 ( ) → #35 9 → #51 + → #43 1 → #42 Shift → #29) → #41 × → #29 ( ) → #43 1 → #48 0 → #48 0 → #46 − → #43 1 → #42 Shift → #29) → #42 Shift → #30 Eval

19. 192 (display `192`)

#29 ( ) → #43 1 → #43 1 → #51 + → #43 1 → #45 3 → #42 Shift → #29) → #41 × → #29 ( ) → #43 1 → #33 7 → #46 − → #35 9 → #42 Shift → #29) → #42 Shift → #30 Eval

20. 625 (display `625`)

#44 2 → #39 5 → #27 x^2 → #42 Shift → #30 Eval

21. 49 (display `49`)

#33 7 → #27 x^2 → #42 Shift → #30 Eval

22. 2.25 (display `2.25`)

#43 1 → #49 . → #39 5 → #27 x^2 → #42 Shift → #30 Eval

23. 361 (display `361`)

#43 1 → #35 9 → #27 x^2 → #42 Shift → #30 Eval

24. 1089 (display `1089`)

#45 3 → #45 3 → #27 x^2 → #42 Shift → #30 Eval

25. 0.25 (display `0.25`)

#48 0 → #49 . → #39 5 → #27 x^2 → #42 Shift → #30 Eval

26. 15625 (display `15625`)

#43 1 → #44 2 → #39 5 → #27 x^2 → #42 Shift → #30 Eval

27. 4096 (display `4096`)

#40 6 → #38 4 → #27 x^2 → #42 Shift → #30 Eval

28. 25 (display `25`)

#42 Shift → #27 $\sqrt{}$ → #40 6 → #44 2 → #39 5 → #42 Shift → #30 Eval

29. 32 (display `32`)

#42 Shift → #27 $\sqrt{}$ → #43 1 → #48 0 → #44 2 → #38 4 → #42 Shift → #30 Eval

30. 99 (display `99`)

#42 Shift → #27 $\sqrt{}$ → #35 9 → #34 8 → #48 0 → #43 1 → #42 Shift → #30 Eval

31. 243 (display `243`)

#45 3 → #21 x^y → #39 5 → #42 Shift → #30 Eval

32. 1.157625 (display `1.157625`)

#43 1 → #49 . → #48 0 → #39 5 → #21 x^y → #45 3 → #42 Shift → #30 Eval

33. 625 (display `625`)

#39 5 → #21 x^y → #38 4 → #42 Shift → #30 Eval

34. 343 (display `343`)

#33 7 → #21 x^y → #45 3 → #42 Shift → #30 Eval

35. 2.25 (display `2.25`)

#43 1 → #49 . → #39 5 → #21 x^y → #44 2 → #42 Shift → #30 Eval

36. 1000000.000 (display `1000000`)

#43 1 → #48 0 → #21 x^y → #40 6 → #42 Shift → #30 Eval

37. 1.41421356 (display `1.414213562`)

#44 2 → #21 x^y → #48 0 → #49 . → #39 5 → #42 Shift → #30 Eval

38. 5 (display `5`)

#48 0 → #49 . → #44 2 → #42 Shift → #36 x^{-1} → #42 Shift → #30 Eval

39. 2 (display `2`)

#26 LOG → #43 1 → #48 0 → #48 0 → #42 Shift → #29) → #42 Shift → #30 Eval

40. 4 (display `4`)

#26 LOG → #43 1 → #48 0 → #48 0 → #48 0 → #48 0 → #42 Shift → #29) → #42 Shift → #30 Eval

41. 2.30258509 (display `2.302585093`)

#25 LN → #43 1 → #48 0 → #42 Shift → #29) → #42 Shift → #30 Eval

42. 0.69314718 (display `0.6931471806`)

#25 LN → #44 2 → #42 Shift → #29) → #42 Shift → #30 Eval

43. -13 (display `−13`)

#28 +/− → #44 2 → #48 0 → #51 + → #33 7 → #42 Shift → #30 Eval

44. 0 (display `0`)

#28 +/− → #45 3 → #51 + → #45 3 → #42 Shift → #30 Eval

45. 150 (display `150`)

#28 +/− → #43 1 → #48 0 → #48 0 → #51 + → #44 2 → #39 5 → #48 0 → #42 Shift → #30 Eval

46. -7 (display `-7`)

#28 +/− → #34 8 → #51 + → #43 1 → #42 Shift → #30 Eval

47. 8000 (display `8000`)

#44 2 → #32 EEX → #45 3 → #41 × → #38 4 → #42 Shift → #30 Eval

48. 30000 (display `30000`)

#43 1 → #49 . → #39 5 → #32 EEX → #38 4 → #41 × → #44 2 → #42 Shift → #30 Eval

49. 900000 (display `900000`)

#45 3 → #32 EEX → #39 5 → #41 × → #45 3 → #42 Shift → #30 Eval

50. 900 (display `900`)

#38 4 → #49 . → #39 5 → #32 EEX → #44 2 → #41 × → #44 2 → #42 Shift → #30 Eval

51. 4.52389342 (display `4.523893421`)

#42 Shift → #45 π → #41 × → #43 1 → #49 . → #44 2 → #27 x^2 → #42 Shift → #30 Eval

52. 10 (display `10`)

#43 1 → #39 5 → #41 × → #40 6 → #42 Shift → #30 Eval → #36 ÷ → #35 9 → #42 Shift → #30 Eval

53. 20 (display `20`)

`#33` 7 → `#49` . → `#39` 5 → `#41` × → `#34` 8 → `#42` Shift → `#30` Eval → `#36` ÷ → `#45` 3 → `#42` Shift → `#30` Eval

54. 30 (display `30`)

`#44` 2 → `#39` 5 → `#48` 0 → `#41` × → `#43` 1 → `#44` 2 → `#42` Shift → `#30` Eval → `#36` ÷ → `#43` 1 → `#48` 0 → `#48` 0 → `#42` Shift → `#30` Eval

55. 15 (display `15`)

`#29` ( ) → `#43` 1 → `#44` 2 → `#51` + → `#43` 1 → `#39` 5 → `#51` + → `#43` 1 → `#34` 8 → `#42` Shift → `#29`) → `#36` ÷ → `#45` 3 → `#42` Shift → `#30` Eval

56. 82 (display `82`)

`#29` ( ) → `#33` 7 → `#48` 0 → `#51` + → `#34` 8 → `#39` 5 → `#51` + → `#35` 9 → `#43` 1 → `#42` Shift → `#29`) → `#36` ÷ → `#45` 3 → `#42` Shift → `#30` Eval

57. 4 (display `4`)

`#29` ( ) → `#44` 2 → `#49` . → `#39` 5 → `#51` + → `#45` 3 → `#49` . → `#39` 5 → `#51` + → `#40` 6 → `#42` Shift → `#29`) → `#36` ÷ → `#45` 3 → `#42` Shift → `#30` Eval

58. 300 (display 300)

#29 ( ) → #43 1 → #48 0 → #48 0 → #51 + → #44 2 → #48 0 → #48 0 → #51 + → #40 6 → #48 0 → #48 0 → #42 Shift → #29) → #36 ÷ → #45 3 → #42 Shift → #30 Eval

59. 73 (display 73)

#43 1 → #48 0 → #48 0 → #46 − → #45 3 → #41 × → #43 1 → #44 2 → #51 + → #35 9 → #42 Shift → #30 Eval

60. 48 (display 48)

#43 1 → #34 8 → #46 − → #44 2 → #41 × → #39 5 → #51 + → #38 4 → #48 0 → #42 Shift → #30 Eval

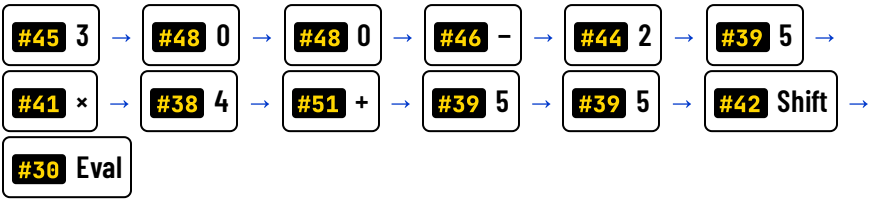
61. 5 (display 5)

#33 7 → #49 . → #39 5 → #46 − → #43 1 → #49 . → #39 5 → #41 × → #45 3 → #51 + → #44 2 → #42 Shift → #30 Eval

62. 42 (display 42)

#40 6 → #38 4 → #46 − → #34 8 → #41 × → #38 4 → #51 + → #43 1 → #48 0 → #42 Shift → #30 Eval

63. 255 (display `255`)



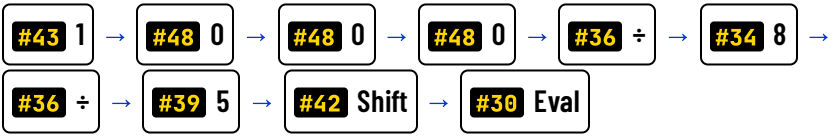
64. 1 (display `1`)



65. 35 (display `35`)



66. 25 (display `25`)



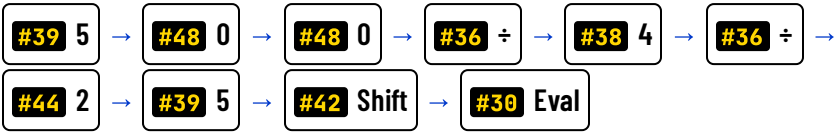
67. 16 (display `16`)



68. 2 (display `2`)



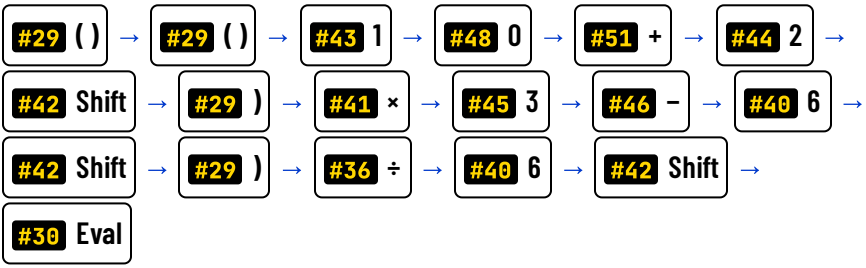
69. 5 (display `5`)



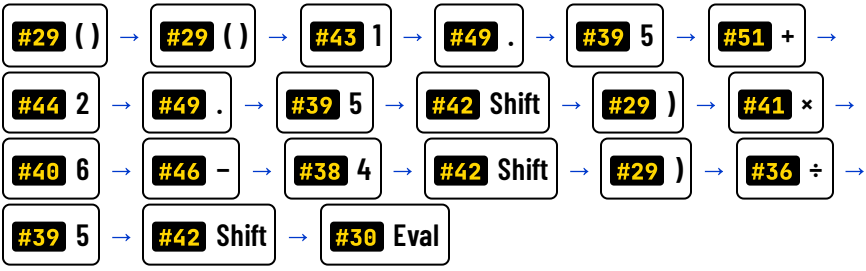
70. 9 (display `9`)



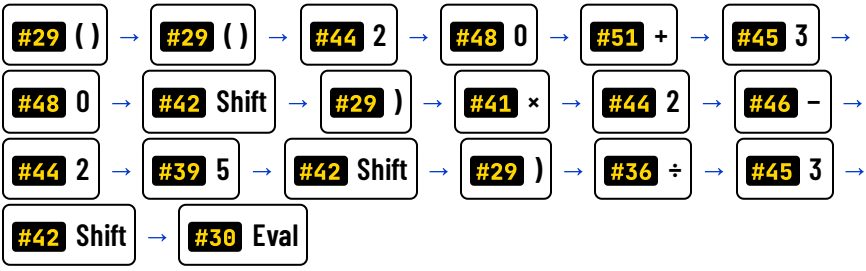
71. 5 (display `5`)



72. 4 (display `4`)



73. 25 (display `25`)



74. 10 (display `10`)

`#29` ( ) → `#29` ( ) → `#33` 7 → `#51` + → `#34` 8 → `#42` Shift → `#29`) → `#41` × → `#35` 9 → `#46` − → `#43` 1 → `#39` 5 → `#42` Shift → `#29`) → `#36` ÷ → `#43` 1 → `#44` 2 → `#42` Shift → `#30` Eval

75. 24 (display `24`)

`#43` 1 → `#44` 2 → `#51` + → `#45` 3 → `#41` × → `#38` 4 → `#42` Shift → `#30` Eval

76. 5 (display `5`)

`#29` ( ) → `#43` 1 → `#44` 2 → `#51` + → `#34` 8 → `#42` Shift → `#29`) → `#36` ÷ → `#38` 4 → `#42` Shift → `#30` Eval

77. 49 (display `49`)

`#29` ( ) → `#45` 3 → `#51` + → `#38` 4 → `#42` Shift → `#29`) → `#41` × → `#29` ( ) → `#35` 9 → `#46` − → `#44` 2 → `#42` Shift → `#29`) → `#42` Shift → `#30` Eval

78. 144 (display `144`)

`#43` 1 → `#44` 2 → `#27` x^2 → `#42` Shift → `#30` Eval

79. 12 (display `12`)

`#42` Shift → `#27` $\sqrt{}$ → `#43` 1 → `#38` 4 → `#38` 4 → `#42` Shift → `#30` Eval

80. 1024 (display `1024`)

#44 2 → #21 x^y → #43 1 → #48 0 → #42 Shift → #30 Eval

81. 4 (display `4`)

#48 0 → #49 . → #44 2 → #39 5 → #42 Shift → #36 x^{-1} → #42 Shift → #30 Eval

82. 1 (display `1`)

#23 COS → #48 0 → #42 Shift → #29) → #42 Shift → #30 Eval

83. 3 (display `3`)

#26 LOG → #43 1 → #48 0 → #48 0 → #48 0 → #42 Shift → #29) → #42 Shift → #30 Eval

84. 0 (display `0`)

#25 LN → #43 1 → #42 Shift → #29) → #42 Shift → #30 Eval

85. 7 (display `7`)

#28 +/− → #39 5 → #51 + → #43 1 → #44 2 → #42 Shift → #30 Eval

86. 1800 (display `1800`)

#40 6 → #32 EEX → #44 2 → #41 × → #45 3 → #42 Shift → #30 Eval

87. 19.63495408 (display `19.63495408`)

#42 Shift → #45 π → #41 × → #44 2 → #49 . → #39 5 → #27 x^2 → #42 Shift → #30 Eval

88. 200 (display `200`)

#43 1 → #48 0 → #48 0 → #41 × → #38 4 → #42 Shift → #30 Eval → #36 ÷ → #44 2 → #42 Shift → #30 Eval

89. 9 (display `9`)

#29 ( ) → #38 4 → #51 + → #34 8 → #51 + → #43 1 → #39 5 → #42 Shift → #29) → #36 ÷ → #45 3 → #42 Shift → #30 Eval

90. 33 (display `33`)

#39 5 → #48 0 → #46 − → #38 4 → #41 × → #40 6 → #51 + → #33 7 → #42 Shift → #30 Eval

91. 10 (display `10`)

#45 3 → #40 6 → #48 0 → #36 ÷ → #38 4 → #36 ÷ → #35 9 → #42 Shift → #30 Eval

92. 3 (display `3`)

#29 ( ) → #29 ( ) → #45 3 → #51 + → #39 5 → #42 Shift → #29) → #41 × → #38 4 → #46 − → #44 2 → #42 Shift → #29) → #36 ÷ → #43 1 → #48 0 → #42 Shift → #30 Eval

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #36
34. #37
35. #41
36. #42
37. #46
38. #47
39. #49
40. #51
41. #42 Shift then #1 Apps
42. #42 Shift then #2 Symb
43. #42 Shift then #8 Esc
44. #42 Shift then #9 ⌂
45. #42 Shift then #10 Plot
46. #42 Shift then #11 View
47. #42 Shift then #12 CAS
48. #42 Shift then #13 Num
49. #42 Shift then #14 Menu
50. #42 Shift then #15 Vars
51. #37 ALPHA then #15 Vars
52. #42 Shift then #16 Toolbox
53. #37 ALPHA then #16 Toolbox
54. #42 Shift then #17 ▭▭
55. #37 ALPHA then #17 ▭▭
56. #42 Shift then #18 xt θn
57. #37 ALPHA then #18 xt θn
58. #37 ALPHA then #19 a/b
59. #42 Shift then #20 ⌫
60. #42 Shift then #21 xy
61. #37 ALPHA then #21 xy
62. #42 Shift then #22 SIN
63. #37 ALPHA then #22 SIN
64. #42 Shift then #23 COS
65. #37 ALPHA then #23 COS
66. #42 Shift then #24 TAN
67. #37 ALPHA then #24 TAN
68. #42 Shift then #25 LN
69. #37 ALPHA then #25 LN
70. #42 Shift then #26 LOG
71. #37 ALPHA then #26 LOG
72. #42 Shift then #27 x^2
73. #37 ALPHA then #27 x^2
74. #42 Shift then #28 +/-
75. #37 ALPHA then #28 +/-
76. #42 Shift then #29 ( )
77. #37 ALPHA then #29 ( )
78. #42 Shift then #30 ,

79. #37 ALPHA then #30 ,
80. #42 Shift then #31 Enter
81. #42 Shift then #32 EEX
82. #37 ALPHA then #32 EEX
83. #42 Shift then #33 7
84. #37 ALPHA then #33 7
85. #42 Shift then #34 8
86. #37 ALPHA then #34 8
87. #42 Shift then #35 9
88. #37 ALPHA then #35 9
89. #42 Shift then #36 ÷
90. #37 ALPHA then #36 ÷
91. #42 Shift then #38 4
92. #37 ALPHA then #38 4
93. #42 Shift then #39 5
94. #37 ALPHA then #39 5
95. #42 Shift then #40 6
96. #37 ALPHA then #40 6
97. #37 ALPHA then #41 ×
98. #42 Shift then #43 1
99. #37 ALPHA then #43 1
100. #42 Shift then #44 2
101. #37 ALPHA then #44 2
102. #42 Shift then #45 3
103. #42 Shift then #46 -
104. #42 Shift then #47 On
105. #42 Shift then #48 0
106. #42 Shift then #49 .
107. #42 Shift then #51 +

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23	EEX	#32
▶Decimal — show the result as a decimal	≈	#42 → #31
10^x — antilog	10x	#42 → #26
Absolute value |x|	|x|	#42 → #28
Addition +	+	#51
ALPHA — the next key types its letter/variable (red/green label)	ALPHA	#37
Ans — recall the last answer into the current expression	Ans	#42 → #51
APPS — applications	Apps	#1
BASE-N — number-base menu (DEC/HEX/BIN/OCT)	Base	#42 → #46
CATALOG — list of all commands	Chars	#42 → #15
CLEAR — clear the entry line / screen	Clear	#42 → #8
CLEAR — clear the entry line / screen	Clear	#42 → #20
Closing bracket). Closing brackets at the end of a line may be omitted	)	#42 → #29
Comma / argument separator (between function arguments, list items or coordinates)	,	#30
Common logarithm $\log_{10}$	LOG	#26
Conversion menu	Units	#42 → #17

Cosine (current angle unit)	COS	#23
Cursor ▲ — up / previous history line / previous item	▲	#3
Cursor ▶ — move right in the line	▶	#6
Cursor ▼ — down / next history line / next item	▼	#4
Cursor ◀ — move left in the line	◀	#5
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	.	#49
DEL — delete the character at the cursor	⌫	#20
Digit 0	0	#48
Digit 1	1	#43
Digit 2	2	#44
Digit 3	3	#45
Digit 4	4	#38
Digit 5	5	#39
Digit 6	6	#40
Digit 7	7	#33
Digit 8	8	#34
Digit 9	9	#35
Division ÷	÷	#36
ENTRY — recall a previous entry	Paste	#42 → #14
Equals — complete the calculation	=	#42 → #49
e^x — exponential	ex	#42 → #25
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	Eval	#42 → #30
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	Enter	#31
Fraction template ▭/▭ (numerator over denominator)	▭▭	#17

Graph / app mode	Settings	#42 → #9
Graph / app mode	Settings	#42 → #12
HOME — return to the main calculation screen	⌂	#9
Imaginary unit i (complex numbers)	i	#42 → #44
Inverse cosine $\cos^{-1}$	ACOS	#42 → #23
Inverse sine $\sin^{-1}$	ASIN	#42 → #22
Inverse tangent $\tan^{-1}$	ATAN	#42 → #24
LIST menu	List	#42 → #33
MATH menu — extra math functions	Menu	#14
MATH menu — extra math functions	Toolbox	#16
MATRIX menu	Matrix	#42 → #35
MATRIX menu	Matrix	#42 → #38
Multiplication ×	×	#41
n-th root $\sqrt[x]{y}$	y√x	#42 → #21
Natural logarithm ln	LN	#25
Negative sign (−) — makes the next number negative. Different from the subtraction key	+/−	#28
Numeric view	Num	#13
OFF — power off (state is lost; press ON to restart)	Off	#42 → #47
ON / AC — power on, or clear all	On	#47
Opening bracket (	( )	#29

Plot view	Plot	#10
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	xy	#21
PRGM — programs	Program	#42 → #43
Printed label only (no action in this app)	␣	#50
QUIT — leave the current screen / app	Esc	#8
Reciprocal x^{-1} (1/x)	x^{-1}	#42 → #36
RESET — return to factory defaults	Mem	#42 → #16
S⇔D — toggle the result between exact (fraction, √, π) and decimal	a⁄b	#19
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	Shift	#42
Shows the message "CAS not available: numeric Home only" (feature not provided in this app)	CAS	#12
Shows the message "Help is not available" (feature not provided in this app)	Help	#7
Shows the message "Info" (feature not provided in this app)	Info	#42 → #1
Shows the message "Notes not available" (feature not provided in this app)	Notes	#42 → #48
Sine (uses the current angle unit)	SIN	#22
Square root √	√	#42 → #27
Square x^2	x^2	#27
STO▶ (→) — store into a variable	Sto ▸	#42 → #32
Subtraction –	–	#46
Symbolic view	Symb	#2
Tangent (current angle unit)	TAN	#24

TBLSET — table start and step	Setup	#42 → #2
TBLSET — table start and step	Setup	#42 → #13
The graphing variable (X / x,θ,t,n) for Y= functions, tables and solving	xt θn	#18
Types "["	[ ]	#42 → #39
Types "{"	{ }	#42 → #34
Types "≠"	≤≥≠	#42 → #40
Variable A (store with STO/ →, recall by name)	A	#37 → #15
Variable B (store with STO/ →, recall by name)	B	#37 → #16
Variable C (store with STO/ →, recall by name)	C	#37 → #17
Variable D (store with STO/ →, recall by name)	D	#37 → #18
Variable E (store with STO/ →, recall by name)	E	#37 → #19
Variable F (store with STO/ →, recall by name)	F	#37 → #21
Variable G (store with STO/ →, recall by name)	G	#37 → #22
Variable H (store with STO/ →, recall by name)	H	#37 → #23
Variable I (store with STO/ →, recall by name)	I	#37 → #24
Variable J (store with STO/ →, recall by name)	J	#37 → #25

Variable K (store with STO/→, recall by name)	K	#37 → #26
Variable L (store with STO/→, recall by name)	L	#37 → #27
Variable M (store with STO/→, recall by name)	M	#37 → #28
Variable N (store with STO/→, recall by name)	N	#37 → #29
Variable O (store with STO/→, recall by name)	O	#37 → #30
Variable P (store with STO/→, recall by name)	P	#37 → #32
Variable Q (store with STO/→, recall by name)	Q	#37 → #33
Variable R (store with STO/→, recall by name)	R	#37 → #34
Variable S (store with STO/→, recall by name)	S	#37 → #35
Variable T (store with STO/→, recall by name)	T	#37 → #36
Variable U (store with STO/→, recall by name)	U	#37 → #38
Variable V (store with STO/→, recall by name)	V	#37 → #39
Variable W (store with STO/→, recall by name)	W	#37 → #40
Variable X (store with STO/→, recall by name)	X	#37 → #41
Variable Y (store with STO/→, recall by name)	Y	#37 → #43
Variable Z (store with STO/→, recall by name)	Z	#37 → #44

VARS — variables menu	Vars	#15
WINDOW — set the graph view	Setup	#42 → #10
WINDOW — set the graph view	Setup	#42 → #11
Y= — enter functions to graph	Define	#42 → #18
ZOOM — zoom presets	View	#11
π (3.14159265…) as an exact constant	π	#42 → #45

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

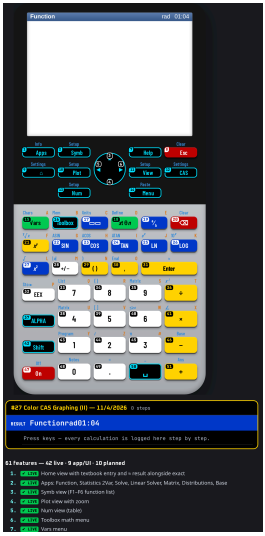
Colour graphing model with Home view (textbook entry, ≈ result alongside exact), Apps (Function, Statistics 2Var, Solve, Linear Solver, Matrix, Distributions, Base), Symb (function list F1–F6), Plot, Num (table), View (zoom), Toolbox (math menu) and Vars. The CAS view, Help, Notes and programming are not simulated (CAS key shows a notice).

Planned (not yet working)

- CAS view
- Help system
- Notes
- Programming
- Geometry app
- Graph image export (PNG)
- Shareable function list link
- Dark screen theme
- Speech read-out of values
- Exam mode

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 Apps ⇧ Info	**#2** Symb ⇧ Setup	**#3** ▲
#4 ▼	**#5** ◀	**#6** ▶
#7 Help	**#8** Esc ⇧ Clear	**#9** ⌂ ⇧ Settings
#10 Plot ⇧ Setup	**#11** View ⇧ Setup	**#12** CAS ⇧ Settings
#13 Num ⇧ Setup	**#14** Menu ⇧ Paste	**#15** Vars ⇧ Chars
#16 Toolbox ⇧ Mem	**#17** ▭▭ ⇧ Units	**#18** xt θn ⇧ Define
#19 a⁄b	**#20** ⌫ ⇧ Clear	**#21** xy ⇧ y√x
#22 SIN ⇧ ASIN	**#23** COS ⇧ ACOS	**#24** TAN ⇧ ATAN
#25 LN ⇧ ex	**#26** LOG ⇧ 10x	**#27** x^2 ⇧ √
#28 +/− ⇧ |x|	**#29** ( ) ⇧)	**#30** , ⇧ Eval
#31 Enter ⇧ ≈	**#32** EEX ⇧ Sto ▸	**#33** 7 ⇧ List
#34 8 ⇧ { }	**#35** 9 ⇧ Matrix	**#36** ÷ ⇧ x^{-1}
#37 ALPHA	**#38** 4 ⇧ Matrix	**#39** 5 ⇧ [ ]
#40 6 ⇧ ≤≥≠	**#41** × ⇧ ∠	**#42** Shift
#43 1 ⇧ Program	**#44** 2 ⇧ i	**#45** 3 ⇧ π
#46 − ⇧ Base	**#47** On ⇧ Off	**#48** 0 ⇧ Notes
#49 . ⇧ =	**#50** ␣ ⇧ _	**#51** + ⇧ Ans

U-45 CALCULATORS · Calculator #27 · cut out and keep